

REV2026

SEMESTER I

THEORY

4 CREDITS

# Mathematics for Computer Technology

Course Code 2001C · L:T:P:S 3:1:0:0 ·  
60 Contact Hours

|               |                                                       |
|---------------|-------------------------------------------------------|
| Programme     | AI, ML, Cloud, Cyber Forensics, CS, IT, Robotics etc. |
| Pre-requisite | 1002 Fundamentals of Engineering Mathematics I        |
| CIE           | 40                                                    |
| ESE           | 60                                                    |
| Self-Learning | 60 hours                                              |

# Official Source Declaration

**Course:** Mathematics for Computer Technology (2001C) · **Curriculum:** Kerala Polytechnic Revision 2026.

This handbook is prepared from the official syllabus PDF. All topics, subtopics, learning outcomes, and assessment rules are faithfully reproduced.

**Verified Inconsistencies:** The PDF lists 4 modules but Module I, II, III have 11 instructional hours each; Module IV has 12 hours (total 45 L + 15 T = 60). The self-learning table suggests 60 hours across 16 weeks. No contradictions detected.

## Course Dashboard

**Programme(s):** Artificial Intelligence, ML, Cloud Computing, Cyber Forensics, CS, IT, Automation, Robotics, etc.

**Subject Code:** 2001C | **Semester:** I | **Type:** Theory

**Teaching Scheme:** 3:1:0:0 (L:T:P:S) | **Contact Hours:** 60

**Credits:** 4 | **CIE:** 40 | **ESE:** 60

### Pre-requisite

| Topic                                   | Course Code | Course Title                            | Semester |
|-----------------------------------------|-------------|-----------------------------------------|----------|
| Trigonometry, Differentiation, Matrices | 1002        | Fundamentals of Engineering Mathematics | I        |

### Teaching and Assessment Scheme

| Teaching/Week | Self-learning/Week | Total Hours/Sem | Credits | CIE | ESE | Total |
|---------------|--------------------|-----------------|---------|-----|-----|-------|
| 3L+1T         | 5.5                | 60              | 4       | 40  | 60  | 100   |

## How to Use This Handbook

**1. Understand** Read the module topics and deep explanations.

**2. Visualise** Study diagrams and interactive calculators.

**3. Apply** Solve worked examples and practice questions.

**4. Revise** Use quick revision cards and glossary before exams.

## Course Outcomes & CO-PO Mapping

### Course Outcomes (CO)

| CO  | Course Outcome                                                                | Bloom's Level |
|-----|-------------------------------------------------------------------------------|---------------|
| CO1 | Apply concepts of complex numbers to solve problems.                          | Apply         |
| CO2 | Use integration techniques to solve basic engineering problems.               | Apply         |
| CO3 | Perform data analysis using the foundation of logic and statistical tools.    | Apply         |
| CO4 | Apply basic graph theory concepts to represent and solve real-world problems. | Apply         |

### CO-PO Articulation Matrix

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO2 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO3 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO4 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -    | -    |

Legends: 1-Low, 2-Medium, 3-High, - No Correlation

## Curriculum Coverage Map

### Module-wise topics, hours, CO, and anchor

| Module | Title             | Topics                                                                                         | Hours | CO  | Anchor    |
|--------|-------------------|------------------------------------------------------------------------------------------------|-------|-----|-----------|
| I      | Complex Numbers   | System, operations, modulus, amplitude, polar form                                             | 11    | CO1 | #module-1 |
| II     | Integral Calculus | Reverse of differentiation, standard results, by parts, substitution, definite integrals, area | 11    | CO2 | #module-2 |

|     |                          |                                                                                                |    |     |           |
|-----|--------------------------|------------------------------------------------------------------------------------------------|----|-----|-----------|
| III | Logic & Basic Statistics | Propositions, logical ops, truth tables, tautology, statistics, grouped data, central tendency | 11 | CO3 | #module-3 |
| IV  | Graph Theory             | Basic concepts, paths, walks, cycles, connectivity, trees, Dijkstra, Kruskal                   | 12 | CO4 | #module-4 |

## Module Roadmap

### Module I

Complex Numbers → algebraic operations, polar form, Argand plane.

### Module II

Integral Calculus → integration as anti-derivative, standard integrals, substitution, by parts, definite integrals.

### Module III

Logic & Statistics → propositions, truth tables, tautology, grouped frequency, mean/median/mode.

### Module IV

Graph Theory → vertices, edges, paths, cycles, trees, Dijkstra shortest path, Kruskal MST.

## MODULE I Complex Numbers

11 Hours

CO1

Apply

Complex numbers extend the real number system to solve equations like  $x^2 + 1 = 0$ . They are essential in AC circuit analysis, signal processing, and control systems.

### The Complex Number System, Conjugate, Equality

A complex number is expressed as  $z = a + ib$ , where  $a$  is the real part ( $\text{Re } z$ ) and  $b$  is the imaginary part ( $\text{Im } z$ ), and  $i = \sqrt{-1}$ .

**Conjugate:** The conjugate of  $z = a+ib$  is  $\bar{z} = a - ib$ .

**Equality:** Two complex numbers are equal iff their real parts are equal and imaginary parts are equal.

**Example:** If  $2x + 3iy = 4 + 6i$ , then  $2x = 4 \rightarrow x = 2$ ;  $3y = 6 \rightarrow y = 2$ .

**Malayalam support:** കോംപ്ലക്സ് സംഖ്യകൾ (complex numbers) (real), (imaginary) സംഖ്യകൾ.  $i = \sqrt{-1}$ .

### Fundamental Operations (+, -, ×, ÷)

- **Addition:**  $(a+ib) + (c+id) = (a+c) + i(b+d)$
- **Subtraction:**  $(a+ib) - (c+id) = (a-c) + i(b-d)$
- **Multiplication:**  $(a+ib)(c+id) = (ac - bd) + i(ad + bc)$
- **Division:**  $(a+ib)/(c+id) = ((a+ib)(c-id))/(c^2+d^2)$

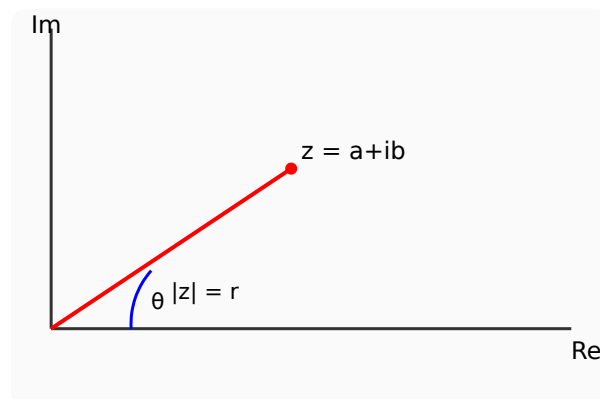
**Example:**  $(3+2i)(1-4i) = 3 \cdot 1 - 2 \cdot (-4) + i(3 \cdot -4 + 2 \cdot 1) = 3+8 + i(-12+2) = 11 - 10i$ .

### Graphical Representation (Argand Plane), Modulus & Amplitude

Complex numbers are represented as points  $(a,b)$  in the Argand plane ( $x = \text{real}$ ,  $y = \text{imaginary}$ ).

**Modulus (Absolute value):**  $|z| = \sqrt{a^2 + b^2}$ .

**Amplitude (Argument):**  $\theta = \tan^{-1}(b/a)$ , measured in degrees or radians.



Argand plane: point  $(a,b)$  with modulus  $r$  and argument  $\theta$ .

### Complex Number Modulus & Argument

Real (a):

Imag (b):

**Modulus: 5.00, Argument: 53.13°**

### Polar Form and Engineering Problems

The polar form of  $z = a + ib$  is  $\mathbf{z = r(cos \theta + i sin \theta)}$ , where  $r = |z|$  and  $\theta = \arg(z)$ .

**Example:** Express  $z = 1 + i$  in polar form.  $r = \sqrt{(1+1)}=\sqrt{2}$ ,  $\theta = \tan^{-1}(1/1)=45^\circ$ .  
So  $z = \sqrt{2} (\cos 45^\circ + i \sin 45^\circ)$ .

**Engineering Application:** Used in AC circuits for phasor representation, control systems, and signal processing.

## MODULE II Integral Calculus

11 Hours

CO2

Apply

Integration is the reverse process of differentiation. It is used to find areas, volumes, and accumulated quantities in engineering.

### Integration as Reverse of Differentiation, Standard Integrals

If  $d/dx [F(x)] = f(x)$ , then  $\int f(x) dx = F(x) + C$ .

$$\int x^n dx = x^{n+1}/(n+1) + C \quad (n \neq -1)$$

$$\int e^x dx = e^x + C$$

$$\int 1/x dx = \ln|x| + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

## Rules of Integration, Substitution ( $f(ax+b)$ , $f(x) \cdot f'(x)$ , $f'(x)/f(x)$ )

- **Constant multiple:**  $\int k f(x) dx = k \int f(x) dx$
- **Sum/Difference:**  $\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$
- **Substitution:** For  $f(ax+b)$ , substitute  $u = ax+b$ ,  $du = a dx$ .
- For  $f(x) \cdot f'(x)$ , let  $u = f(x)$ ,  $du = f'(x) dx$ .
- For  $f'(x)/f(x)$ , result is  $\ln|f(x)| + C$ .

**Example:**  $\int (2x+3)^5 dx$ . Let  $u = 2x+3$ ,  $du = 2 dx \rightarrow dx = du/2$ .  $= 1/2 \int u^5 du = (1/2)(u^6/6) = (2x+3)^6/12 + C$ .

## Integration by Parts

**Formula:**  $\int u dv = u v - \int v du$ . Choose  $u$  as the function that simplifies on differentiation (LIATE rule).

**Example:**  $\int x \sin x dx$ . Let  $u = x$ ,  $dv = \sin x dx \rightarrow du = dx$ ,  $v = -\cos x$ .  $= -x \cos x - \int (-\cos x) dx = -x \cos x + \sin x + C$ .

## Definite Integrals and Area Under Curve

**Definite integral:**  $\int_a^b f(x) dx = F(b) - F(a)$ .

**Area bounded by curve and x-axis:**  $A = \int_a^b |f(x)| dx$ .

**Example:**  $\int_0^1 x^2 dx = [x^3/3]_0^1 = 1/3$ .

### Definite Integral Calculator ( $x^2$ )

a:

b:

**Area = 0.3333**

Propositions and truth tables form the basis of digital logic. Statistics helps in data analysis and decision making.

### Propositions, Logical Operations (Negation, Conjunction, Disjunction, XOR)

**Proposition:** A statement that is either true or false.

- **Negation ( $\neg p$ ):** Opposite truth value.
- **Conjunction ( $p \wedge q$ ):** True only if both are true.
- **Disjunction ( $p \vee q$ ):** True if at least one is true.
- **Exclusive OR ( $p \oplus q$ ):** True if exactly one is true.

Truth Table for Basic Operations

| p | q | $\neg p$ | $p \wedge q$ | $p \vee q$ | $p \oplus q$ |
|---|---|----------|--------------|------------|--------------|
| T | T | F        | T            | T          | F            |
| T | F | F        | F            | T          | T            |
| F | T | T        | F            | T          | T            |
| F | F | T        | F            | F          | F            |

### Tautology, Contradiction, Contingency

- **Tautology:** Always true (e.g.,  $p \vee \neg p$ ).
- **Contradiction:** Always false (e.g.,  $p \wedge \neg p$ ).
- **Contingency:** Neither always true nor false.

### Statistics: Grouped Data, Frequency Distribution, Measures of Central Tendency

**Grouped data** is organised into classes. **Frequency distribution** shows how often each value occurs.

**Mean:**  $\bar{x} = (\sum f x) / \sum f$

**Median:** Middle value when data is ordered.

**Mode:** Most frequent value.



**Example:** Data: 2,3,5,5,7,8. Mean =  $(2+3+5+5+7+8)/6 = 5.0$ . Median =  $(5+5)/2 = 5$ . Mode = 5.

## MODULE IV Graph Theory

12 Hours

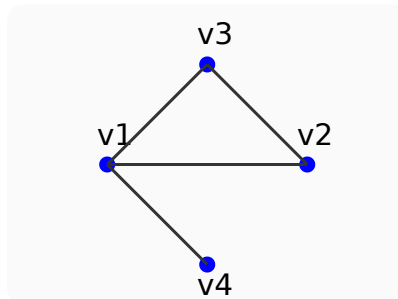
CO4

Apply

Graphs model networks, relationships, and paths. Used in computer networks, circuit design, and route planning.

### Basic Concepts: Vertex, Edge, Degree, Loops, Simple Graph

- **Vertex:** Node or point.
- **Edge:** Line connecting two vertices.
- **Degree:** Number of edges incident to a vertex.
- **Loop:** Edge from a vertex to itself.
- **Simple graph:** No loops or multiple edges.



Simple graph with vertices v1-v4 and edges.

### Handshaking Lemma, Subgraphs, Union of Graphs

**Handshaking Lemma:** Sum of degrees =  $2 \times$  (number of edges).

**Subgraph:** A graph formed from a subset of vertices and edges.

**Union:** Combined graph of two graphs with common vertices.

### Matrix Representation: Adjacency and Incidence Matrices

**Adjacency matrix:** A  $n \times n$  matrix where  $A[i][j] = 1$  if edge exists, else 0.

**Incidence matrix:**  $n \times m$  matrix where  $B[i][j] = 1$  if vertex  $i$  is incident to edge  $j$ .

### Types of Graphs: Null, Complete, Path, Cycle, Connected, Regular, Weighted, Tree, Spanning Tree

- **Null:** No edges.
- **Complete ( $K_n$ ):** Every pair of vertices connected.
- **Path:** Linear sequence of vertices and edges.
- **Cycle:** Closed path.
- **Connected:** Path between every pair.
- **Regular:** All vertices have same degree.
- **Weighted:** Edges have weights.
- **Tree:** Connected acyclic graph.
- **Spanning tree:** Subgraph that is a tree and includes all vertices.

### Dijkstra's Algorithm (Shortest Path) & Kruskal's Algorithm (Minimum Spanning Tree)

**Dijkstra:** Finds shortest path from source to all vertices in weighted graph.

**Kruskal:** Finds minimum spanning tree by sorting edges by weight and adding if no cycle.

**Example (Kruskal):** Edges: AB(2), AC(3), BC(5), BD(1). Sorted: BD(1), AB(2), AC(3), BC(5). Add BD, AB, AC  $\rightarrow$  MST weight =  $1+2+3=6$ .

## Formula Bank

**Complex:**  $z = a+ib$ ,  $|z| = \sqrt{a^2+b^2}$ ,  $\arg z = \tan^{-1}(b/a)$ , polar =  $r(\cos\theta + i \sin\theta)$

**Integration:**  $\int x^n dx = x^{n+1}/(n+1)$ ,  $\int e^x dx = e^x$ ,  $\int 1/x dx = \ln|x|$

**By Parts:**  $\int u dv = uv - \int v du$

**Definite:**  $\int_a^b f(x) dx = F(b) - F(a)$

**Logic:**  $\neg(p \wedge q) = \neg p \vee \neg q$ ,  $p \vee \neg p = T$

**Statistics:** Mean =  $\Sigma fx / \Sigma f$ , Median = middle value, Mode = most frequent

**Graph:** Sum of degrees =  $2|E|$ , Dijkstra, Kruskal MST

## Visual Library

Argand Plane — complex number representation

Integration Area — area under curve

Graph Model — vertices, edges, trees

## Self-Learning Activities

**Module I:** Find conjugate, modulus, amplitude, polar form of complex numbers. (16 hours)

**Module II:** Evaluate integrals using substitution and by parts; apply definite integrals. (13 hours)

**Module III:** Construct truth tables; solve mean/median/mode problems; applications of statistics. (14 hours)

**Module IV:** Draw real-life graphs; identify paths/cycles; find applications of graph theory. (17 hours)

## Assessment Methodology

### CIE and ESE Breakdown

| Mode      | Attendance            | CA1 | CA2 | CA3 | CA4 | CA5 | CA6 | ESE |
|-----------|-----------------------|-----|-----|-----|-----|-----|-----|-----|
| Marks     | 5                     | 15  | 15  | 15  | 15  | 50  | 60  | 60  |
| Converted | 5                     | 15  |     |     | 20  |     |     | 60  |
| Total     | CIE 40 + ESE 60 = 100 |     |     |     |     |     |     |     |

**Series Examination Pattern:** 2 hours, any one module. Part A ( $2 \times 3 = 6$ ), Part B ( $3 \times 3 = 9$ ), Part C ( $5 \times 4 = 20$ ) — total 35 marks.

## Module-wise Questions with Answers

### Module I

**1. Define complex number and find the conjugate of  $3+4i$ .**

A complex number  $z = a+ib$ . Conjugate =  $3-4i$ .

**2. Express  $z = 1 - i$  in polar form.**

$r=\sqrt{2}$ ,  $\theta=-45^\circ$ ,  $z=\sqrt{2}(\cos 45^\circ - i \sin 45^\circ)$ .

### Module II

**3. Evaluate  $\int (3x^2+2x) dx$ .**

$x^3 + x^2 + C$ .

**4. Find area under  $y = x^2$  from  $x=0$  to  $2$ .**

$\int_0^2 x^2 dx = [x^3/3]_0^2 = 8/3$ .

### Module III

**5. Construct truth table for  $p \wedge q$ .**

$T \& T \rightarrow T$ ,  $T \& F \rightarrow F$ ,  $F \& T \rightarrow F$ ,  $F \& F \rightarrow F$ .

**6. Find mean of data: 2,4,6,8,10.**

Mean = 6.

### Module IV

### 7. What is a tree? Give example.

A connected acyclic graph. Example: family tree.

### 8. State Handshaking Lemma.

Sum of degrees =  $2 \times$  number of edges.

## Practice Model Paper

*Based on the official pattern, not an official SITTTTR model question paper.*

#### Part A (2 marks each)

1. Define complex number. 2. What is the conjugate of  $5-2i$ ? 3. Write the integral of  $1/x$ . 4. Define tautology.

#### Part B (3 marks each)

5. Express  $2+2i$  in polar form. 6. Evaluate  $\int x \sin x \, dx$ . 7. Construct truth table for  $p \vee q$ .

#### Part C (5 marks each, with choice)

8. a) Find modulus and argument of  $-1+i$ . OR b) Evaluate  $\int_0^1 (x^2+x) \, dx$ . 9. a) Apply Kruskal's algorithm to find MST. OR b) Explain Dijkstra's algorithm.

## Quick Revision

**Module I**  $i^2=-1$ ,  $|z|=\sqrt{a^2+b^2}$ , polar  $r(\cos\theta+i \sin\theta)$

**Module II**  $\int x^n \, dx = x^{n+1}/(n+1)$ , by parts:  $\int u \, dv = uv - \int v \, du$

**Module III** Truth tables, mean= $\Sigma fx/\Sigma f$ , median=middle, mode=most frequent

**Module IV** Graph =  $(V,E)$ , tree = connected acyclic, MST via Kruskal

**Common Mistakes:** Forgetting  $+C$  in indefinite integrals; mixing real/imaginary parts; incorrect truth table for XOR.

**Exam Checklist:** Revise formulas, practice truth tables, solve at least one MST and shortest path problem.

## Glossary

| Term           | Meaning                                |
|----------------|----------------------------------------|
| Complex Number | Number of form $a+ib$                  |
| Modulus        | Absolute value, distance from origin   |
| Amplitude      | Angle of complex number with real axis |
| Integral       | Anti-derivative, area under curve      |
| Proposition    | Statement with truth value             |
| Tautology      | Always true                            |
| Contradiction  | Always false                           |
| Graph          | Set of vertices and edges              |
| Tree           | Connected acyclic graph                |
| Spanning Tree  | Tree containing all vertices           |

## Learning Resources

**Textbook:** Mathematics for Computer Technology, SITTTTR Kalamassery.

**Web Resources:** (None listed in PDF)